



BHARATIYA ANTARIKSH HACKATHON 2025

Powered by **IIS**



Team Name : Inevitables

Team Leader Name : Akshit Kotnala

Problem Statement : Simulation/Modelling of Forest Fire Spread using AI/ML techniques

Team Members

Team Leader:

Name: Akshit Kotnala

College: Army Institute of Technology Pune

Team Member-1:

Name: Shivam Kapasiya

College: Army Institute of Technology Pune

Team Member-2:

Name: Vishal Ghosh

College: Army Institute of Technology Pune

Team Member-3:

Name:

College:

Brief about the Idea:

Forest fires are growing more frequent and severe due to climate change, rising temperatures, and human encroachment. These fires threaten biodiversity, human lives, and property, and they often spread faster than authorities can respond. Most existing systems detect fires after they've already begun, offering little time for preventive action. To address this, our idea is to build **an AI/ML-powered forest fire spread simulation model that can predict how a fire might evolve before it causes irreversible damage.**

Our system will use a **combination of satellite imagery, weather data** (such as wind speed, humidity, and temperature), **vegetation indices** (like NDVI), and **topographic features** to forecast the likely direction, speed, and extent of a wildfire. Through spatiotemporal modeling techniques, the platform will simulate the fire's behavior over time, helping authorities visualize its potential spread in the coming hours or days.

Users will be able to select a region or input geographic coordinates through an intuitive interface. The tool will then display an interactive map showing predicted fire zones, allowing decision-makers to prepare accordingly. This includes planning evacuations, prioritizing vulnerable zones, and efficiently deploying firefighting resources.

By **combining geospatial data with machine learning and simulation**, we aim to **shift wildfire management from reactive to proactive.** Our goal is to build a scalable, accessible system that supports real-time forecasting and better disaster response—ultimately minimizing loss to life, nature, and economy.

● How different is it from any of the other existing ideas?

- Predicts **fire spread**, not just detects fire.
- Uses **multi-source data** (satellite, weather, NDVI, topography).
- Provides **interactive map-based simulation**.
- Allows **custom location input** for localized forecasting.

● How will it be able to solve the problem?

- Enables **early planning and evacuation**.
- Supports **smart deployment** of firefighting resources.
- Reduces damage through **data-driven forecasts**.
- Enhances **situational awareness** with visual predictions.

● USP of the proposed solution

A proactive, AI-powered wildfire simulation engine that visualizes the future path of forest fires before they ignite chaos.

- **Predictive simulation**, not just detection.
- **Multi-source AI/ML modeling** for higher accuracy.
- **User-friendly, geo-coordinate based simulation interface.**
- Designed for **scalability**—can be used in different regions with minimal setup.
- **Supports disaster planning** and saves lives, ecosystems, and infrastructure.

List of features offered by the solution

AI/ML-Powered Forest Fire Spread Simulation

Forest fires are growing more frequent and severe due to climate change, rising temperatures, and human encroachment. These fires threaten biodiversity, human lives, and property, spreading faster than authorities can respond.

Our system simulates how a fire might evolve, forecasting the likely direction, speed, and extent using satellite imagery, weather data, vegetation indices, and topographic features



Satellite Imagery



Weather Data



Vegetation Indices

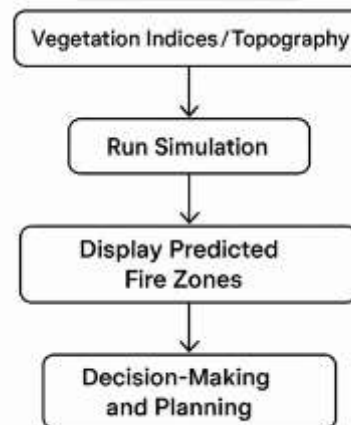
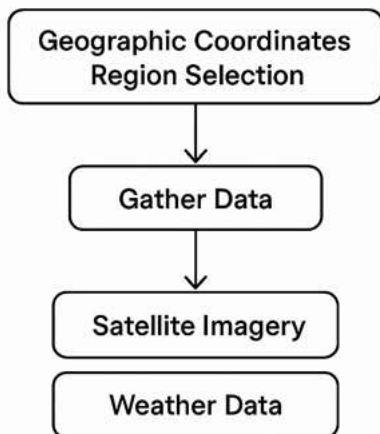


Topographic Features



Process flow diagram Or Use-case diagram

Forest Fire Spread Simulation Using AI/ML



Wireframes/Mock diagrams of the proposed solution (optional)

Forest Fire Spread Simulation

Geographic Coordinates

Predicted Fire Zones

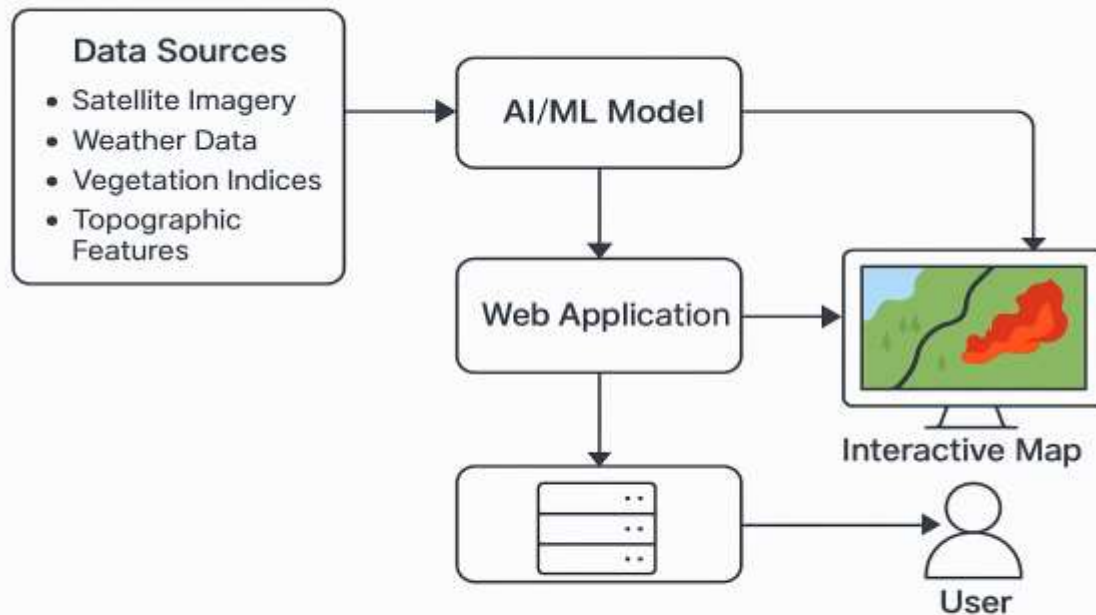
+

-



Decision-Making and Planning

Architecture diagram of the proposed solution



Architecture Diagram of the AI/ML-Enhanced Wildfire Prediction System

Technologies to be used in the solution:

1 Programming & Machine Learning Frameworks

- **Python** – Core programming language.
- **Scikit-learn** – For traditional ML models.
- **TensorFlow / PyTorch** – For deep learning, particularly spatiotemporal models.

2. Spatiotemporal Modeling

- **ConvLSTM** – For capturing spatial and temporal fire spread patterns.
- **Temporal Fusion Transformers (TFT)** – For sequence forecasting with interpretability.
- **Other Sequence-based ML Models** – Suitable for dynamic predictions over time and space.

3. Satellite & Environmental Data Sources

- **Google Earth Engine** – For accessing and processing satellite data.
- **MODIS, Sentinel-2, INSAT** – Satellite imagery for vegetation, temperature, and fire detection.
- **ERA5, NASA POWER** – Reanalysis weather datasets for wind speed, humidity, and temperature.

4. Geospatial Tools

- **GeoPandas** – For geospatial vector data manipulation.
- **Rasterio** – For reading and writing raster datasets.
- **QGIS** – For advanced spatial data visualization and analysis.
- **Vegetation Indices (e.g., NDVI)** – To assess fire-prone vegetation areas.

Estimated implementation cost (optional):

Category	Item	Estimated Cost (USD)
Data APIs	Weather & Satellite APIs (Free Tiers)	\$0 – \$100
Cloud Compute	Model Training (GPU on GCP/AWS)	\$100 – \$300
Web Hosting	App & Backend Deployment	\$30 – \$100
Storage	Cloud Storage for Data	\$20 – \$50
Visualization Tools	Mapbox / Leaflet (Free)	\$0
Miscellaneous	Domain, Testing Tools	\$20 – \$50

BHARATIYA ANTARIKSH HACKATHON 2025

THANK YOU

